```
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# CS 82A - Data Science
# module8_lab.ipynb
# penguins.csv
# Instructor Seno
# 10/14/2026
```

Step 1: Open Colab and load the penguins

Create a Colab notebook named module8_lab.ipynb, upload penguins.csv (Tutorial 1), and load it with pandas.

You should see: a DataFrame with 344 rows and columns including species, flipper_length_mm, body_mass_g

```
from google.colab import files
files.upload()

import pandas as pd
df = pd.read_csv('penguins.csv')
```

Choose Files penguins.csv

penguins.csv(text/csv) - 16735 bytes, last modified: 10/14/2026 - 100% done

```
Saving penguins.csv to penguins (3).csv
```

```
df.head()
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mas
0	Adelie	Torgersen	39.1	18.7	181.0	37
1	Adelie	Torgersen	39.5	17.4	186.0	38
2	Adelie	Torgersen	40.3	18.0	195.0	32
3	Adelie	Torgersen	NaN	NaN	NaN	
4	Adelie	Torgersen	36.7	19.3	193.0	34

Next steps: Generate code with df | New interactive sheet

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 344 entries, 0 to 343
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   species            344 non-null    object
 1   island             344 non-null    object
 2   bill_length_mm     342 non-null    float64
 3   bill_depth_mm      342 non-null    float64
 4   flipper_length_mm  342 non-null    float64
 5   body_mass_g        342 non-null    float64
 6   sex                333 non-null    object
 7   year               344 non-null    int64
dtypes: float64(4), int64(1), object(3)
memory usage: 21.6+ KB
```

A DataFrame with 344 rows and columns including species, flipper_length_mm, body_mass_g.

Step 2: Establish ground truth by hand, no AI

Record in code and markdown: the shape, the missing-value counts per column, and the number of penguins of each species.

Hint: three lines you have used since Module 2, plus value_counts().

You should see: a handful of missing values in the measurement columns; three species with known counts. This is your truth table for Step 5.

```
# 1. Dataset Shape
print("Dataset Shape:")
df.shape
# Output: (344, 8) -> 344 rows, 8 columns
```

```
Dataset Shape:
(344, 8)
```

```
df.isnull().sum()
# Output confirms missing counts: - bill_length_mm (2), - bill_depth_mm (2), -
```

	0
species	0
island	0
bill_length_mm	2
bill_depth_mm	2
flipper_length_mm	2
body_mass_g	2
sex	11
year	0

dtype: int64

```
# 3. Number of penguins of each species
print("Penguins Per Species:\n")
print(df["species"].value_counts())
```

```
Penguins Per Species:

species
Adelie       152
Gentoo       124
Chinstrap     68
Name: count, dtype: int64
```

Species counts:

- Adelie: 152,
- Gentoo: 124,
- Chinstrap: 68.

Step 3: Prompt the AI (and log it)

Fill in the Tutorial 3 prompt template with this dataset's real columns and dtypes, send it to your AI assistant, and paste the FULL prompt into a markdown cell titled Prompt Log.

Prompt Log

1. Full AI Prompt Sent:
2. The problem with the AI is that it pasted istelf into the upper top cells to run, with the scope of the csv file below it. In this case re-running the notebook from top to

bottom may produce errors.

3. Second problem is that a code snippet that I was using was dropped and deleted without my permission. Saving the ipynb file before AI use might be crucial.
4. The nice thing about the AI is that it's well organized and neat. It's very readable and sets a nice example.
5. I had to add a print statement for the AI output, number three.
6. The groubpby function returning flipper length mean of each penguin species, matches exactly.
7. (Step 6 follow-up - One Judgement)An important aspect of data display is the 4 second takeaway presentation. By sharing what the data is revealing the reader can understand more quickly. Such as "Gentoo penguin flipper length is longer than Chinstrap and Adelle penguins.". This might be a better title.

You are helping with exploratory data analysis. My CSV has columns and dtypes:

- species (object)
- island (object)
- bill_length_mm (float64)
- bill_depth_mm (float64)
- flipper_length_mm (float64)
- body_mass_g (float64)
- sex (object)
- year (int64)

Write Python code using pandas and seaborn to: (1) summarize missing data, (2) compare flipper_length_mm across species by computing the group means, (3) produce one labeled chart showing this distribution. Explain each step in comments.

Step 4: Run the AI's code

Paste the returned code into fresh cells and run it. If something errors, fix it and add one line to the prompt log describing the fix.

You should see: a missing-data summary, a species comparison, and at least one chart.

```
# Import necessary libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# (1) Summarize missing data
# We compute the count of missing values per column to understand completeness
```

```
missing_summary = df.isnull().sum()
print("--- (1) Missing Data Summary ---")
print(missing_summary)

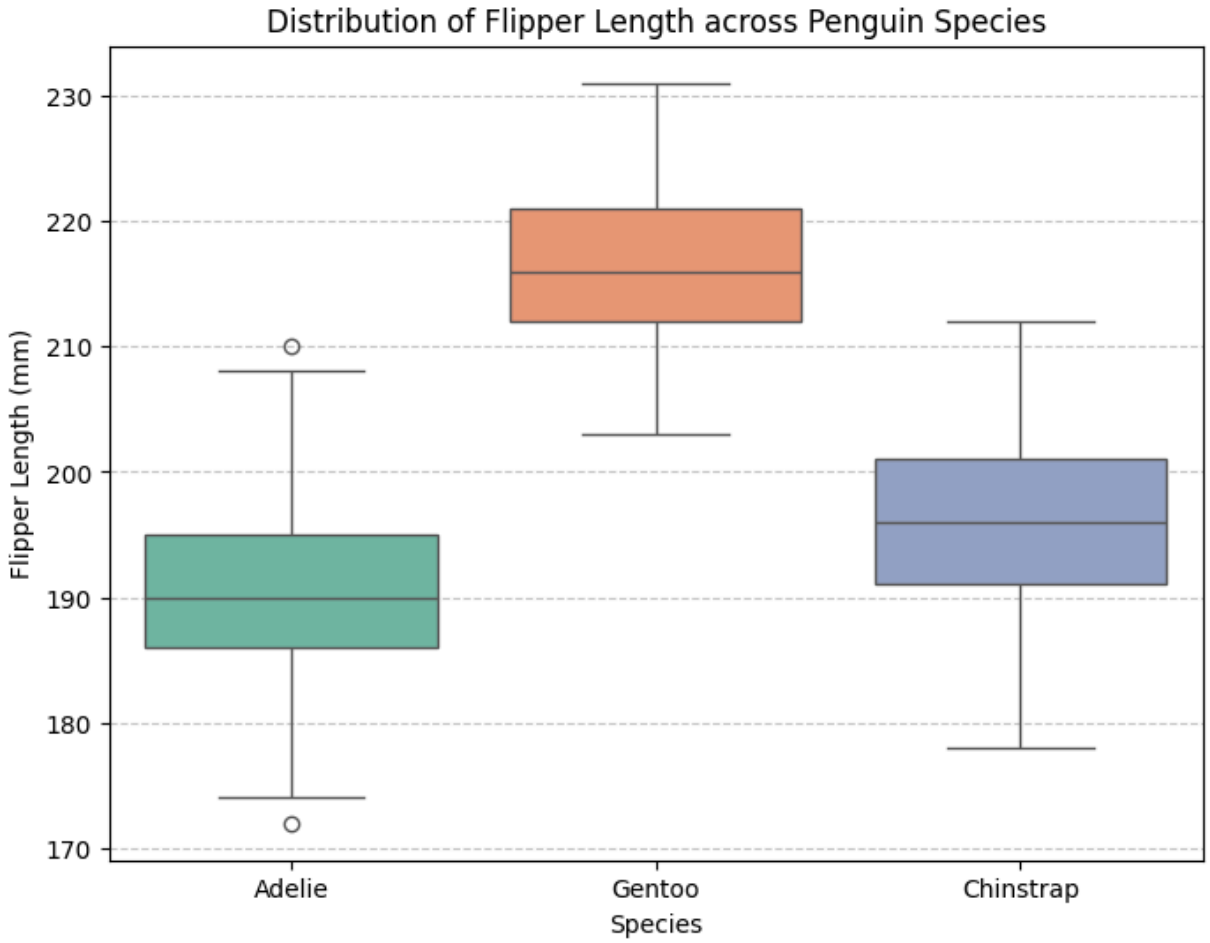
# (2) Compare flipper_length_mm across species by computing the group means
# We group the dataset by species and find the mean flipper length
species_flipper_means = df.groupby('species')['flipper_length_mm'].mean()
print("\n--- (2) Group Means for Flipper Length (mm) ---")
print(species_flipper_means)

# (3) Produce one labeled chart showing this distribution
# We use a boxplot to display the distribution of flipper_length_mm across the
print("\n--- (3) Display the distribution of flipper_length_mm across the diffe
plt.figure(figsize=(8, 6))
sns.boxplot(x='species', y='flipper_length_mm', hue='species', data=df, palette
plt.title('Distribution of Flipper Length across Penguin Species')
plt.xlabel('Species')
plt.ylabel('Flipper Length (mm)')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

```
--- (1) Missing Data Summary ---
species               0
island                0
bill_length_mm        2
bill_depth_mm         2
flipper_length_mm     2
body_mass_g           2
sex                  11
year                  0
dtype: int64

--- (2) Group Means for Flipper Length (mm) ---
species
Adelie       189.953642
Chinstrap    195.823529
Gentoo       217.186992
Name: flipper_length_mm, dtype: float64

--- (3) Display the distribution of flipper_length_mm across the different speci
```



Step 5: The verification checkpoint

Compute the mean flipper length per species YOURSELF with one groupby line (Tutorial 2 pattern). Compare against any per-species numbers the AI reported, and record match or mismatch in the log. If they differ, explain why (missing-value handling is the usual culprit).

You should see: either agreement, or a mismatch you can explain. Both earn full credit; silence does not.

```
# Manual verification check
df.groupby("species")["flipper_length_mm"].mean()
```

species	flipper_length_mm
Adelie	189.953642
Chinstrap	195.823529
Gentoo	217.186992

dtype: float64

Verification of numbers matching mine.

Observations: Using my df.groupby(species)[flipper length].mean() function returns the exact same values as the AI because we are using the same csv file and same python code. **Adelie: 189.95** , **Chistrap: 195.82** and **Gentoo: 217.186**. These are the same calculations from the AI.

Adelie 189.953642

Chinstrap 195.823529

Gentoo 217.186992

The calculations that are made both would ignore the NaN values still existing in the dataset.

Step 6: One follow-up, one judgment

Ask the AI one follow-up that improves the analysis (a better chart, an extra grouping). Run it, then keep or reject the suggestion with one sentence of your own judgment in the

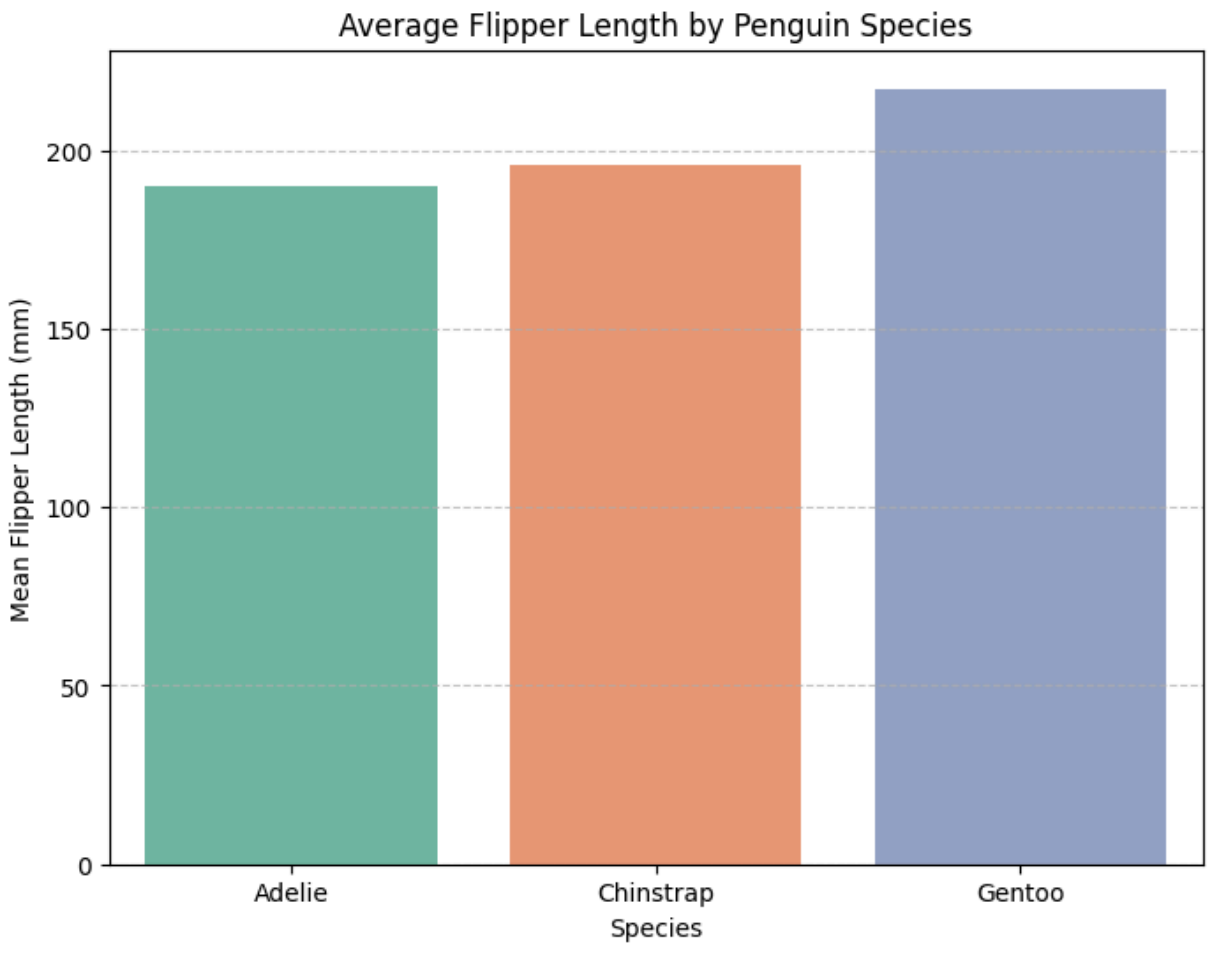
log.

AI prompt: (3) Display the distribution of flipper_length_mm across the different species using a bar plot instead of boxplot.

```
# Import necessary libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Compute mean flipper length per species
species_means = df.groupby('species')['flipper_length_mm'].mean().reset_index()

# Create a bar plot representing the means
plt.figure(figsize=(8, 6))
sns.barplot(x='species', y='flipper_length_mm', hue='species', data=species_mea
plt.title('Average Flipper Length by Penguin Species')
plt.xlabel('Species')
plt.ylabel('Mean Flipper Length (mm)')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



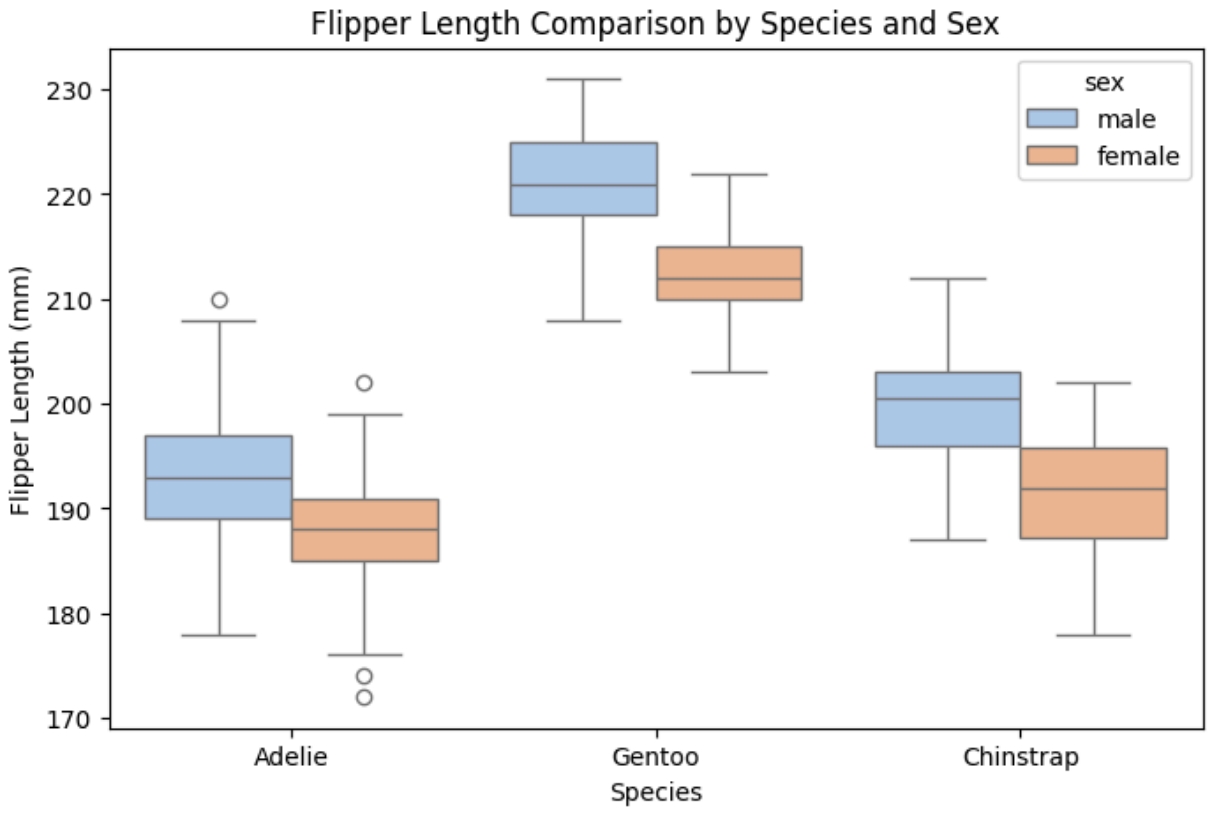
The data is more clear to understand and more familiar looking, with Gentoo above Chinstrap and Adelle traling closely. But it lacks a takeaway that works in 4 seconds.

Step 7: Your original insight, no AI

In markdown, one insight about penguins the AI did not state, backed by a number or chart you produce yourself.

Modify the code to split the flipper length boxplot by sex to see if males and females differ within each species?

```
plt.figure(figsize=(8, 5))
sns.boxplot(data=df.dropna(subset=["sex"]), x="species", y="flipper_length_mm",
plt.title("Flipper Length Comparison by Species and Sex")
plt.xlabel("Species")
plt.ylabel("Flipper Length (mm)")
plt.show()
```



The sex of the penguins shows that they have longer flippers than females.

```
df.groupby("island")["body_mass_g"].mean()
```

island	body_mass_g
Biscoe	4716.017964
Dream	3712.903226
Torgersen	3706.372549

dtype: float64

The groupby function shows that, it's possible that Biscoe island has more food to eat for penguins such as fish and vegetation than the other islands such as Torgersen. The penguins do seem to have a combined heavier weight, on Biscoe island than on other islands. Which could mean that it's preferred by penguins.

Step 8: Privacy reflection (markdown cell)

This dataset is public animals. Name one dataset from your own life or work you must NEVER paste into a public AI tool, and what you would do instead (anonymize, aggregate, or use an approved internal tool). 2–3 sentences.